Fall Creek Rainfall–Runoff Model

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BEE4310 – Environmental Statistics and Learning

Introduction

Being able to predict the abundance and availability of natural resources is becoming increasingly important under ongoing climate change. Future flow predictions for rivers and streams can not only provide stakeholders with insights for risk-informed decision-making, but also support the development of effective mitigation and adaptation strategies in response to changing hydrological regimes.

Recent research on small watersheds, such as the Fall Creek case study by Peng and Albertson (2021)[1], has demonstrated the value of data-driven hydrological frameworks for detecting changes in precipitation–runoff dynamics that are not readily observable from streamflow records alone. By inferring hidden hydrological parameters through Bayesian estimation, their work highlights how local watershed responses can shift over time due to climatic variability or anthropogenic influences. Building on this perspective, the present study adopts a complementary, prediction-oriented approach by combining machine-learning-based flow modelling with climate projection data to assess how future hydroclimatic changes may manifest in both mean streamflow and seasonal flow patterns.

Within this project, two research questions are addressed: *(1) How will Fall Creek's streamflow develop in the near future under a high-carbon emission scenario, and (2) how will these changes affect the creek's flow seasonality?*

Addressing these questions can help communities within the Fall Creek watershed better understand plausible future streamflow conditions and their potential implications for local livelihoods and lifestyles. Key stakeholders include agricultural producers who depend on streamflow for irrigation, for whom changes in flow magnitude and timing may influence land- and water-management decisions. In addition, alterations in discharge extremes and seasonal flow patterns may affect erosion and sedimentation processes, with consequences not only for agriculture but also for ecosystem health and recreational use of the river.

Data and Methods

The rainfall–runoff model was trained using three primary predictors: maximum temperature, minimum temperature, and precipitation. Historical observations of these variables were obtained from the Cornell University Weather Station for the period 1925–2025, with data accessed through the National Oceanic and Atmospheric Administration (NOAA) [2]. Streamflow data covering the same period were collected from a river gauge located upstream of Beebe Lake and obtained from the United States Geological Survey (USGS) National Water Information System [3].

All historical observations were available at a daily temporal resolution and were therefore aggregated to monthly averages. Subsequently, the climate and streamflow datasets were merged into a single dataset based on matching year and month.

The climate predictors were further extended by including monthly lagged values, up to six months for precipitation and three months for temperature. This approach was adopted to increase the model's memory, allowing it to better explain streamflow variability by implicitly accounting for processes such as snowfall accumulation, snow depth, and snowmelt, which are strongly governed by precipitation and temperature. This consideration is particularly important, as the watershed is affected by snow processes for a substantial portion of the year.

Nonparametric regression models were selected for this study because these explain the non-linearity in data reasonably well, since the relationship between the response variable and the predictors is learned from the data, hence no parametric form is assumed. This property is especially valuable in hydrological modeling, where strong seasonality and daily volatility are key charateristics of the system.

Two tree-based ensemble models introduced in the course were trained: a Random Forest Regressor (RF) and a Gradient Boosting Regressor (GB). Random Forest constructs an ensemble of regression trees using bootstrap samples and randomly selected subsets of predictors, and generates predictions by averaging the outputs of all trees. In contrast, Gradient Boosting builds a sequence of weak learners, where each successive tree is trained to reduce the residual errors of the preceding tree, ultimately forming a single strong predictive model.

Further, the models' hyperparameters were optimized using cross-validation within a 70–30% train–test data split. Within the training portion, a validation batch was held out and not used for model fitting. The trained model was then used to predict responses for this validation batch, and the differences between the observed and predicted values were quantified to evaluate model performance. This cross-validation loop went over 5 different validation splits, in searching for the best hyperparameters such as the amount of trees, the depth of the trees, the minimum leaf sample, the learning rate (GB) or how the amount of sampled predictors per tree is chosen (RF).

This cross-validation process captures the relationships within the training data and is performed prior to testing. Consequently, when the model is evaluated on the test set, it has already undergone hyperparameter optimization, enabling it to produce its best possible predictions.

Model selection was primarily conducted through an iterative trial-and-error process, in which both quantitative performance metrics and qualitative visual assessment were considered. Multiple combinations of lagged predictors were evaluated (e.g., a three-month precipitation lag combined with a one-month temperature lag). The final selected model was a Gradient Boosting regressor incorporating six months of precipitation lags and three months of temperature lags.

This model demonstrated satisfactory predictive performance, achieving a root mean squared error (RMSE) of 78 relative to a mean streamflow of 188 cubic feet per second, as well as an R^2 value of 75%, indicating a strong ability to explain variance in the target variable. In addition, the adjusted R^2 metric was used for model comparison to assess whether the inclusion of additional predictors led to genuine improvements in explanatory power rather than gains attributable to chance.

Lastly, bias–variance diagnostics were conducted to guide an informed trial-and-error process when adding or removing predictors, particularly with respect to the number of lagged climate variables included in the model.

Using the Gradient Boosting (GB) model, future streamflow was predicted through 2050 based on the NEX–GDDP–CMIP6 climate projection dataset [4], which is derived from global climate model (GCM) simulations and provided by the NASA Center for Climate Simulation. Projections were obtained under the SSP5-8.5 (SSP585) scenario, which represents a high greenhouse gas emission pathway characterized by continued fossil-fuel–driven economic growth and limited climate mitigation, and is commonly used to assess "worst-case scenarios".

The case study location was specified using the geographic coordinates of Ithaca, New York, namely a latitude of 42.4° N and a longitude of 76.4° W, which corresponds to 283.6° in the 0–360° longitude convention.

Lagged climate predictors were incorporated consistently during the projection period. For the initial months of the forecast horizon, lag values were derived from their corresponding historical observations, while for subsequent months they were progressively replaced by lagged values obtained from the GCM climate projections as the predictions advanced in the future.

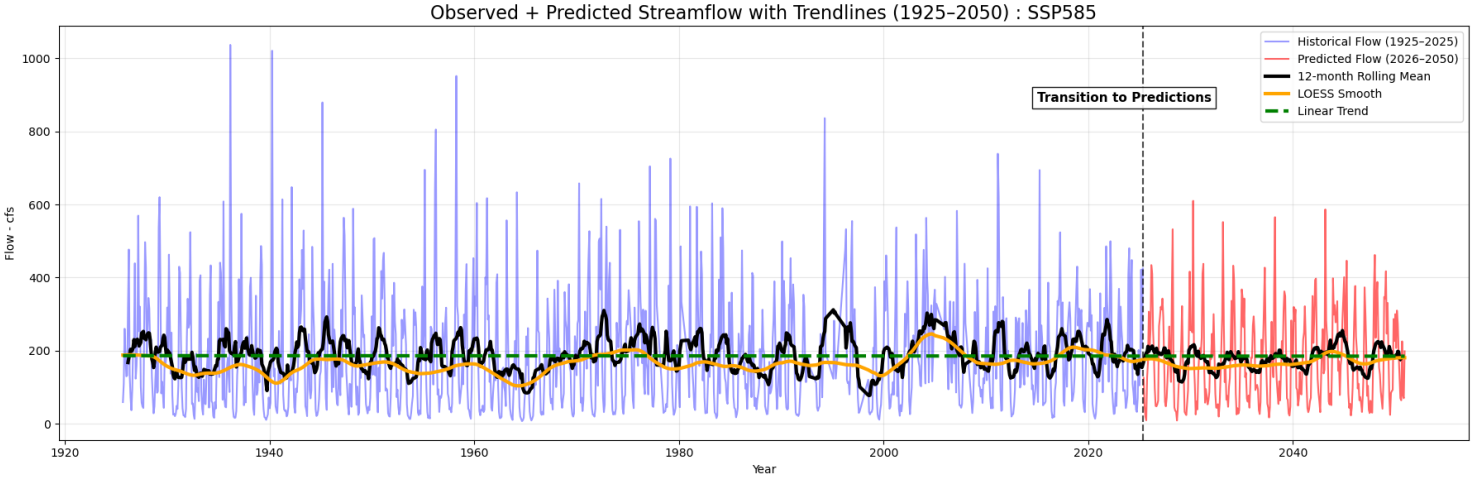

Figure 1: Fall Creek Monthly Streamflow 1925-2050 timeline

Results

Figure 1 presents the observed monthly streamflow of Fall Creek from 1925 to 2025 (blue), followed by projected monthly streamflow from 2026 to 2050 (red). A linear trend fitted to the complete time series exhibits a slightly negative but nearly neutral slope of (-0.001), indicating that the long-term mean streamflow is not projected to change substantially by mid-century under the applied scenario.

Figure 2 (see next page) illustrates the typical annual seasonality of streamflow, comparing a recent historical period with a projected future period under the SSP585 scenario. Monthly mean discharge values were averaged within each period to

construct representative annual hydrographs for Fall Creek.

The comparison reveals pronounced shifts in the seasonal distribution of streamflow. Winter months, particularly December and January, are projected to experience significantly higher mean flows relative to the historical period (e.g., an increase of approximately $110\,\mathrm{ft}^3\,\mathrm{s}^{-1}$ in mean January flow). In contrast, spring and autumn show reduced mean discharge, while summer flows exhibit a modest increase.

The projected increase in winter streamflow is consistent with warmer temperatures under the SSP585 scenario, which are expected to promote a greater fraction of precipitation falling as rain and earlier snowmelt, thereby reducing winter snow storage. Conversely, diminished spring and autumn flows may reflect earlier depletion of snowpack combined with increased evapotranspiration during warmer months.

These seasonal shifts in streamflow timing may have important implications for reservoir operations, agricultural water availability, and ecological processes that depend on the natural flow regime of Fall Creek.

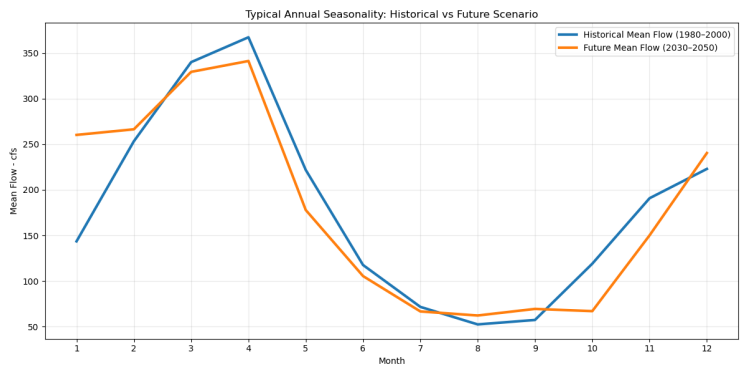

Figure 2: Average seasonal streamflow for a historical period and a projected future period under SSP585.

Discussion and Conclusion

Overall, the results indicate that under the high-emission SSP585 scenario, Fall Creek's mean streamflow is not projected to change substantially by mid-century. However, pronounced changes in the seasonal distribution of flow are evident, with increased winter discharge and reduced flows during spring and autumn. These findings suggest that future hydrological impacts in the watershed may be driven more by changes in seasonality and variability than by shifts in long-term mean streamflow.

As shown in Figure 1, the model captures the long-term behaviour and seasonal signal of the river reasonably well but appears to underrepresent the short-term volatility observed in the historical record when projecting future flows. This limitation likely arises from a combination of factors, including the restricted set of climate predictors, the finite memory introduced through lagged variables, and the use of monthly-averaged data, which smooths short-duration fluctuations. In addition, although tree-based ensemble methods can model nonlinear relationships, the chosen model complexity may still be insufficient to fully reproduce the observed variability in streamflow.

Because the analysis and projections are based on monthly averaged values, the results primarily provide insight into changes in baseflow conditions rather than daily discharge extremes. Extreme flow events, however, are of particular relevance to stakeholders within the watershed. For example, farmers may need to adapt water collection and management practices during periods of unusually high discharge, while recreational activities may face increased safety risks. Such extremes are also of high importance for engineers involved in the design of river infrastructure, as design flows and design storms are typically based on short-duration peak discharges rather than long-term averages. Moreover, elevated peak flows can be associated with land-use changes or saturated soil conditions, which enhance surface runoff and alter groundwater recharge dynamics. Such processes may have broader implications for water supply reliability within the watershed.

In addition, changes in flow extremes may influence nutrient transport and sediment delivery to Cayuga Lake, potentially exacerbating nutrient pollution, eutrophication, and associated biodiversity loss. The inability of the present model to explicitly capture these extreme hydrological events therefore represents an important limitation, particularly when considering environmental risks and water-quality impacts.

Finally, it should be noted that the presented projections are based on a single global climate model and a single emission scenario, and thus represent one plausible future trajectory rather than a probabilistic forecast. While the Gradient Boosting model is effective at capturing nonlinear relationships within the historical training period, uncertainties associated with extrapolation under future climate conditions remain. Future work could address these limitations by incorporating multiple climate models and scenarios, higher temporal resolution data, and additional hydrological or land-use predictors to better constrain uncertainty and improve representation of extreme events.

References

[1] X. Peng and J. D. Albertson. A data-driven diagnostic framework for small watersheds based on markov chain-monte carlo, 2021.

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[4] NASA Center for Climate Simulation. NEX–GDDP–CMIP6 Climate Projection Dataset. NASA Earth Exchange (NEX), 2021. Downscaled CMIP6 climate projections.

Acknowledgements

ChatGPT (OpenAI) was used to assist with language refinement, grammar correction, and improvement of sentence flow.